

Quiz — Nonlinear Equations in One Variable and Systems of Equations in Two Variables

Student: Lavanya Thondavada **Date:** 2026-09-17 **Time limit:** 10 questions × 95 sec ≈ **16 minutes**

Instructions: No notes. Desmos is available for every question — decide for each one whether it's faster to graph or faster to work by hand. Show your work for the fill-in questions in the space provided.

1. [Easy] 332cd67b

$3x^2 - 15x + 18 = 0$. How many distinct real solutions are there to the given equation?

A. Exactly one B. Exactly two C. Infinitely many D. Zero

2. [Easy] Clone of 1e003284

$$x = 64 \quad y = \sqrt{x} + 5$$

The graphs of the given equations intersect at the point (x, y) in the xy -plane. What is the value of y ?

(fill in)

3. [Medium] Clone of 4e18fc5d

$u = -t / (80s)$. The given equation relates the distinct positive numbers u , t , and s . Which equation correctly expresses t in terms of u and s ?

A. $t = -80us$ B. $t = -80u/s$ C. $t = -s/(80u)$ D. $t = u + 80s$

4. [Medium] d0a7871e

$$y = x + 1 \quad y = x^2 + x$$

If (x, y) is a solution to the system of equations above, which of the following could be the value of x ?

A. -1 B. 0 C. 2 D. 3

5. [Medium] a5663025

A system of equations consists of a quadratic equation and a linear equation. The graphs of the equations in this system (shown at right — a parabola and a line crossing at two points) are given in the xy -plane. How many solutions does this system have?

A. 0 B. 1 C. 2 D. 3

6. [Medium] 7f81d0c3

$x^2 - x - 1 = 0$. What values satisfy the equation above?

A. $x = 1$ and $x = 2$ B. $x = -1/2$ and $x = 3/2$ C. $x = (1+\sqrt{5})/2$ and $x = (1-\sqrt{5})/2$ D. $x = (-1+\sqrt{5})/2$ and $x = (-1-\sqrt{5})/2$

7. [Hard] 3a9d60b2

$2|4 - x| + 3|4 - x| = 25$. What is the positive solution to the given equation?

(fill in)

8. [Hard] Clone of ba0edc30

$x^2 - 4x - 5 = 0$. One solution to the given equation can be written as $2 + \sqrt{k}$, where k is a constant. What is the value of k ?

(fill in)

9. [Hard] Clone of fc3d783a

In the xy -plane, a line with equation $3y = 6$ intersects a parabola at exactly one point. If the parabola has equation $y = -2x^2 + bx$, where b is a positive constant, what is the value of b ?

(fill in)

10. [Hard] 4661e2a9

$$x - y = 1 \quad x + y = x^2 - 3$$

Which ordered pair is a solution to the system of equations above?

A. $(1+\sqrt{3}, \sqrt{3})$ B. $(\sqrt{3}, -\sqrt{3})$ C. $(1+\sqrt{5}, \sqrt{5})$ D. $(\sqrt{5}, -1+\sqrt{5})$

Answer Key — Nonlinear Equations Quiz (Lavanya Thondavada, 2026-09-17)

#	Answer	Source	Rationale	Desmos or By hand
1	B	332cd67b	Discriminant = $(-15)^2 - 4(3)(18) = 9 > 0 \rightarrow$ exactly two real solutions.	By hand — discriminant arithmetic is faster than graphing.
2	13	Clone of 1e003284	Substitute $x=64$: $y = \sqrt{64} + 5 = 8 + 5 = 13$.	By hand — direct substitution.
3	A	Clone of 4e18fc5d	Multiply both sides by $-80s$: $-80su = t$, so $t = -80us$.	By hand — one-step algebra.
4	A	d0a7871e	$x+1 = x^2+x \rightarrow x^2 = 1 \rightarrow x = \pm 1$. Only -1 is a choice.	Desmos — graph $y_1=x+1$, $y_2=x^2+x$, click intersections and read x .
5	C	a5663025	The two graphs cross at 2 points $\rightarrow 2$ solutions.	By hand — the graph is already given; just count crossings.
6	C	7f81d0c3	Quadratic formula: $x = (1 \pm \sqrt{5})/2$.	Desmos — graph $y_1=x^2-x-1$, $y_2=0$, click both x -intercepts for decimal values, match to choice C.
7	9	3a9d60b2	$5 4-x =25 \rightarrow 4-x =5 \rightarrow x=-1$ or $x=9$. Positive solution is 9.	By hand — splitting into two cases is quick and exact.
8	9	Clone of ba0edc30	$(x-2)^2 = 9 \rightarrow x = 2 \pm 3$. Matching $2+\sqrt{k}$ form: $\sqrt{k}=3 \rightarrow k=9$.	By hand — completing the square is direct here.
9	4	Clone of fc3d783a	$y=2$; $2x^2-bx+2=0$; discriminant=0 $\rightarrow b^2=16 \rightarrow b=4$.	Desmos — put a slider on b and drag until the line just touches the parabola, then read the slider value.
10	A	4661e2a9	Substitution gives $y^2=3$, $y=\sqrt{3}$, $x=1+\sqrt{3}$.	Desmos — graph both equations, click the intersection, and match the decimal reading to choice A.

Desmos-fastest: #4, #6, #9, #10. By-hand-fastest: #1, #2, #3, #5, #7, #8 — both are represented, per the rule.